Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/30/2010
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Project Title:	The role of mechanoperception in gravity responses of the model organism Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	77005	**Congressional District:**	25
Comments:			
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No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA ARC
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Key Personnel Changes/Previous PI:			
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Task Description:	The fruit fly Drosophila melanogaster offers unique opportunities for genetic and molecular studies of cellular processes common to all multicellular organisms. In order to learn more of the pathways activated by the force of gravity, we have isolated a series of Drosophila mutants that have altered behavioral responses to gravity. These and other Drosophila mutants will be used to explore the signaling pathways underlying one the most marked changes associated with exposure to microgravity - loss of muscle tissue. Specifically the studies will address the issue of whether mechanosensory neurons and associated central brain circuitry regulate muscles responses to altered gravitational force. A powerful technique has been developed within Drosophila that allows conditional inactivation of targeted neurons. This technique will be used to inactivate the neurons under study and to determine whether this loss of neural input alters the responses of the muscles to hypergravity. Quantitative PCR will be used to examine muscle responses. A selection of genes indicated by other studies to show altered expression in response to altered gravitational force will be examined.		

	These studies should provide unique information on the regulation of muscle metabolism in response to changes in gravity. NRA NNH08ZTT003N seeks to identify studies aimed at understanding signaling pathways associated with mechanoperception. The studies proposed are focused directly upon this goal. NASA has programmatic interest in understanding the adaptation of living things to change in gravitational force. The studies presented will provide significant insight into these adaptive processes. The major team member associated with these studies is senior research technician Rebecca A. Simonette.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The studies under this project address the question of how muscles respond to increase in mechanical force by growth or decrease in mechanical force by atrophy. In addition to providing insight into the effects of hypo-and hyper-gravity on muscle mass these studies are highly applicable to an understanding of medical situations that occur on Earth. In particular, the muscle atrophy experience from confinement to bedrest almost certainly involves similar regulatory mechanisms. By uncovering the mechanisms by which muscles sense and respond to altered mechanical this research will have significant impact upon human health.
Task Progress:	Muscles are highly responsive to mechanical load. Muscles that are heavily used undergo growth whereas muscles that are not subjected to mechanical work atrophy and become weak. Degeneration of the muscles is thus a common problem for astronauts under the low gravitational force experienced in space. A major question concerning these observations is – how do the muscles sense their mechanical load and how do they respond to it? One hypothesis is that neurons within the muscles specialized to sense the force exerted on the muscles somehow provide feedback to the muscles initiating either growth or atrophy. The research performed under this project is aimed at investigating this possible relationship between mechanosensory neurons and muscle metabolism. The experiments are being performed in the model organism Drosophila melanogaster, a small fruit fly. This organism has been studied intensively for about 100 years and as a result, many research tools are available for studies of muscle function in this organism that have not been generated for other animal species. The findings from this work should provide insight into a major health problem associated with space flight and thus provide the groundwork for intervention at the molecular level. The current project grew out of our previous work to identify genes that contribute to responses to gravitational force. During the past year we have established that two of these genes are associated with neuromuscular function and have begun investigating how they contribute to muscle activity. One of these genes, which we have called jim lovell or lov (in honor of the Apollo 13 astronaut) has characteristics that suggest its molecular action is to control the specific functions of a particular class of mechanosensory neurons termed chordotonal neurons. We have established during the year that Lov is not expressed in all chordotonal organs but is expressed in several associated with muscles involved in locomotion. Given this association of the Lov-expressing neurons with muscles associated with movement, we set out to determine whether the Lov neurons actually control muscle activity. To investigate this possibility, we used one of the tools especially developed in Drosophila melanogaster. This is a genetic method that allows the researcher to express a particular protein in cell types that do not normally express the protein. The actual experiments performed involved expressing Lov in chordotonal neurons that do not normally express Lov, the rationale being that if Lov controls muscle activity and thus helps produce coordinated muscle movement by being expressed in a particular chordotonal neurons, locomotion should be disrupted in some way if Lov is expressed in different neurons. Initially we chose to apply this approach during the larval stage of development since the nervous system is much better understood during this period. Strikingly, when Lov is misexpressed in this way, the coordinated muscle movements of the larvae are dramatically affected. The larvae develop from embryos in eggshells prepared by their mothers. At the end of embryonic development the newly formed larva must perform a particular series of muscle movements in order to hatch from the eggshell. The larvae with misdirected expression of Lov are so clumsy that many of them are incapable of performing the muscle movements necessary for hatching from their eggshells. The hatched larvae move by a “caterpillar like” locomotion that involves highly controlled waves of contraction along the body from the back to the front. When the Lov misexpressing larvae are physically removed from their eggshells and set on a smooth surface they fail to generate these coordinated waves of contraction and cannot move from any position in which they are placed. Ultimately the larvae starve since they are also incapable of the movements needed for feeding. We have also misexpressed Lov during the adult stage of Drosophila development. The adults form in pupal cases generated by the larvae at the end of larval development. When Lov is misexpressed so as to affect adult locomotor neurons, most adults are incapable of eclosing from their pupal cases, again because of failure to make the necessary coordinated movements. If helped from their pupal cases, the adults are highly uncoordinated, falling frequently, failing to fly and dying rapidly due to an inability to forage effectively for food. These studies establish that the normal function of Lov in the mechanosensory neurons is to provide the muscles with the feedback information necessary to generate coordinated muscle activity. Our next step is to determine if this activity of Lov also affects muscle activity, by examining the expression of genes in the muscles associated with growth. We will also examine the reverse situation and determine whether loss of Lov in the appropriate neurons causes atrophy of the muscles. The second gene that we have undertaken to study, is termed yuri gagarin, or yuri. This gene encodes a protein that appears to interact with a key component of the muscle contraction apparatus. This protein, termed Tropomyosin, has a major role in stabilizing one of the rod-like structures in muscles that move during the contraction process. Using another specialized technique termed TAP-tagging, we have shown that the protein encoded by Yuri binds directly to the Tropomyosin protein. Given that we know that mutations to the yuri gene affect locomotor responses to gravity, these findings suggest that both the protein encoded by the yuri gene and the Tropomyosin protein could be muscle proteins that are regulated by the mechanosensory neurons that express Lov. Investigation of this possibility will be a component of our future research. Summary. These studies take advantage of the unique possibilities for investigating muscle responses to altered gravitational/mechanical force in the organism Drosophila melanogaster. The two genes under study are allowing us to investigate responses from the muscle-associated mechanosensitive neurons to the muscles themselves.

Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Peer-reviewed Journals	Kracklauer MP, Wiora HM, Chen X, Deery WJ, Bolival B Jr, Romanowicz D, Simonette RA, Fuller MT, Fischer JM, Beckingham KM. "The Drosophila SUN protein Dspag4 cooperates with the coiled-coil protein Yuri Gagarin to position the basal body on the spermatid nucleus." Journal of Cell Science, accepted with minor revisions, March 2010. , Mar-2010
Articles in Peer-reviewed Journals	Texada MJ, Deery WJ, Simonette RA, Beckingham KM. "Tropomyosin is a direct interaction partner of the Drosophila coiled coil protein Yuri Gagarin." Molecular Biology of the Cell, submitted, 2010. , Jan-2010